Beyond Keyword Alerts:
Building a Semantic PR Opportunity Monitor for a Real Agency

A Multi-Client News Matching System Using Dense Retrieval and RAG

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1. Introduction and Problem Motivation

This project began with a conversation with a working PR practitioner at a small Arizona agency. The agency manages media relations for several professional service clients: attorneys, healthcare providers, and telecom companies. Currently, their workflow for finding pitch opportunities is almost entirely manual. Their process today includes subscribing to trade journals specific to each client, configuring Google Alerts or Cision keyword notifications, and scanning incoming results each morning.

The fundamental limitation of keyword alerts is that they can only surface stories the PR professional already thought to search for. Adjacent opportunities, stories a client is genuinely qualified to comment on, but where the headline uses different language than the configured alert are missed entirely. A second limitation the agency identified is that keyword matching surfaces stories where the client's name or topic already appears. That is often a story about a competitor or one where the client has already been quoted. What they actually want are stories where their client has not yet been quoted but could be opportunities to insert the client into an ongoing conversation. This is the distinction between reactive monitoring and proactive pitching, and it is exactly what semantic search enables.

When asked what a better tool would look like, the answer was concrete: 'Something that reads the news the way I read the news. I want to know when a story breaks where my client has something interesting to say, even if the story doesn't use their exact keywords.' The agency was open to sharing real client data across four clients spanning personal injury law, telecom, business law, and healthcare. Each client provided years of real media placements and published coverage (not synthetic data) giving the system a genuine, diverse corpus as the foundation for evaluation and all design decisions that followed.

2. System Overview

The system automatically surfaces relevant news stories for PR clients and generates grounded pitch drafts. For each client, we build an "expertise fingerprint" by embedding their historical article corpus into a ChromaDB vector store. Incoming news headlines are matched against this fingerprint using dense semantic retrieval, with cosine similarity scores determining match strength. Strong matches trigger a RAG-generated media pitch ready for the PR practitioner to review and send. The system serves four clients across three industries. The intended operational cadence is weekly: run the pipeline once or twice per week, review strong matches, send approved pitches. Adjacent matches require more PR judgment and are surfaced as secondary opportunities. A web-based digest UI renders all matches by client, displays generated pitches, and captures practitioner ratings that feed into a human evaluation loop for future reranking improvement.

3. Client Data

3.1 Corpus Construction

Training data was collected through a three-stage pipeline. First, media hits documents, a list of article placements provided by the client or agency, were parsed to extract URLs. Second, a custom web scraper using newspaper3k (primary) and BeautifulSoup (fallback) retrieved full article text from those URLs. Third, a Playwright-based browser automation layer recovered articles blocked by bot detection on the first pass. Client 1's corpus is unique in that it consists of written article drafts and final published pieces provided directly by the client rather than scraped from URLs. This produces higher-quality text and is the gold standard corpus in our evaluation.

3.2 Data Cleaning

Raw scraped text contained boilerplate from news site footers: copyright notices, newsletter signup CTAs, and navigation elements. We applied targeted regex patterns to strip these from article endings without modifying real content. 154 of 901 articles were cleaned, and a 100-word minimum threshold filtered out paywall stubs and truncated previews.

3.3 Client 2 Calendar Seed Documents

Client 2's corpus was the smallest (94 articles) and had limited topic diversity. 47 seed documents were written to cover health awareness topics from Client 2's 2026 calendar not represented in their 94 scraped articles. Each document is 60-100 words, written from Client 2's perspective, explicitly referencing their community centers, primary care physicians, and patient population. These anchor the embedding space for topics like National Kidney Month, World Stroke Day, and Suicide Prevention Awareness Month where Client 2 has no prior media coverage.

Client	Industry	Articles	Topics	Date Range	Details
Client 1	Personal Injury Law	125	28	2020-2025	Covers car accidents, DUI, dog bite, pedestrian safety, pool drowning, autonomous vehicles
Client 2	Telecom	613	29	2021-2025	514 of 613 articles labeled 'general (limits topic-level evaluation)
Client 3	Business Law	194	Mixed	2022-2025	Heavy commercial real estate and employment law coverage
Client 4	Healthcare	94 + 47	Health topics	2022-2025	94 scraped + 47 health awareness calendar seeds covering topics not in scraped corpus

4. System Architecture and Methods

4.1 Client Expertise Modeling with Embeddings

To identify whether a news story represents a potential pitch opportunity, the system first needs a representation of what each client is knowledgeable about. We model this expertise using dense vector embeddings derived from the client's historical media coverage. Each article in a client's corpus is encoded using the sentence-transformers implementation of the all-MiniLM-L6-v2 model. This model is a lightweight bi-encoder architecture based on BERT that converts text into a 384-dimensional semantic vector. Articles discussing similar topics produce vectors that are close to one another in embedding space, even when they use different vocabulary.

This representation allows the system to capture semantic similarity rather than simple keyword overlap, which is critical for PR monitoring. Journalistic coverage often describes the same issue using different wording across outlets, making keyword matching unreliable. The model is implemented using the sentence-transformers library with normalize_embeddings=True, which converts dot product comparisons into cosine similarity scores. This normalization allows consistent similarity scoring across queries and documents. The resulting embeddings form what we refer to as a client's "expertise fingerprint", a vector space representation of the topics they have historically commented on.

4.2 Vector Storage and Client Isolation

Once articles are embedded, they must be stored in a system that supports efficient similarity search. We use ChromaDB, a vector database designed for embedding-based retrieval. Each client's corpus is stored in a separate ChromaDB collection. This architectural choice ensures that similarity search is performed only within the context of that client's expertise. If all articles were stored in a shared collection, a query about a car accident could potentially retrieve telecommunications or healthcare articles that share superficial semantic similarities but are irrelevant to the client. Each stored record includes: article text, embedding vector, and metadata (source, topic, outlet tags). This structure allows the system to retrieve both the most relevant articles and the contextual information required for downstream pitch generation.

4.3 Semantic Retrieval and Hybrid Search Design

When new news headlines are collected, the system evaluates whether they align with a client's expertise by comparing their embeddings to the client's corpus vectors. We initially designed the system using a hybrid retrieval approach combining dense semantic similarity with sparse keyword-based matching. Hybrid retrieval can sometimes improve performance by capturing both semantic meaning and exact term matches.

The hybrid score is calculated as:

$$score = \alpha \times dense_similarity + (1 - \alpha) \times sparse_similarity$$

where dense similarity comes from cosine similarity between embeddings and sparse similarity is computed using TF-IDF features.

The rationale for including sparse retrieval was that some domains, particularly legal or medical fields, contain specialized terminology where exact keyword matches could provide additional signal. However, empirical evaluation using a leave-one-out retrieval consistency metric showed that dense retrieval alone consistently outperformed hybrid retrieval across all clients. This result reflects the diversity of journalistic language in the corpus: the same issue may be described with widely varying vocabulary across publications. In these cases, semantic embeddings capture relevance more effectively than keyword overlap. As a result, the system operates with $\alpha = 1.0$, effectively using dense retrieval only, although the hybrid framework remains available for future clients whose corpora may benefit from sparse signals.

4.4 News Ingestion Pipeline

To identify potential pitch opportunities, the system gathers new news headlines each time the pipeline is run. These headlines serve as the query input for the semantic retrieval system. News stories are retrieved using Google News RSS feeds, which provide a free and rate-limit-free source of current headlines. Queries are configured separately for each client using topic-relevant keywords, such as accident-related terms for the personal injury client or broadband infrastructure terms for the telecommunications client.

Early versions of the system used highly specific queries to limit irrelevant results. However, this approach produced very low headline volumes. The final design instead uses broader queries combined with downstream semantic filtering. This strategy allows the retrieval model to determine which headlines are relevant rather than relying solely on keyword specificity. Headlines are deduplicated within each run and filtered to remove press release wire services, which often produce high semantic similarity scores but do not represent viable pitch opportunities.

Per weekly run, the system fetches approximately:

Client	Headlines per Run	Queries Used	Strong Matches (typical)
Client 1	15-25	10 topic-specific RSS queries	3-10 (varies by news cycle)
Client 2	25-35	4 RSS queries including direct brand name	0-5 (telecom news cycle dependent)
Client 3	20-30	6 business/legal RSS queries	2-8
Client 4	30-45	8 health + awareness month queries	2-8 (higher during awareness months)

4.5 Retrieval-Augmented Pitch Generation

When a news headline produces a strong similarity match with a client's expertise corpus, the system generates a draft media pitch using a retrieval-augmented generation (RAG) approach. In this stage, the system retrieves the top-K most similar articles from the client's corpus. These retrieved articles provide context about the client's previous commentary on related topics. The news headline and retrieved context are then passed to a large language model to generate a pitch that connects the client's past expertise to the current story.

We use K = 3 retrieved articles as using too few articles risks weak contextual grounding, while too many can introduce irrelevant information that degrades generation quality. Pitch generation is performed using a Llama model accessed through the Groq API (llama-3.3-70b-versatile), with prompt instructions that require the model to ground claims in retrieved coverage and avoid generic PR language. This retrieval grounding ensures that generated pitches are not speculative but instead reference real expertise the client has previously demonstrated in the media.

4.6 Match Scoring and Opportunity Thresholds

Not every semantic match represents a viable PR opportunity. To help practitioners focus on the most promising stories, similarity scores are categorized into three tiers:

- Strong matches (≥ 0.55)
- Adjacent matches (0.40-0.55)
- No match (< 0.40)

Only strong matches trigger automatic pitch generation and up to five strong matches are surfaced per client in each weekly digest. Adjacent matches are included in the weekly digest as possible opportunities requiring additional human judgment. The strong-match threshold was raised from an initial value of 0.50 after evaluation revealed that some unrelated Arizona news stories, particularly sports coverage, were scoring above the original threshold due to superficial semantic overlap. Increasing the threshold significantly reduced these false positives.

5. Evaluation

The PR Opportunity Monitor was evaluated using a combination of a quantitative retrieval consistency metric and human judgment of pitch relevance, reflecting both the system's technical performance and its real-world utility.

5.1 Consistency (Retrieval Metric)

To measure how effectively the system identifies relevant news stories, we evaluated the semantic retrieval component using a leave-one-out retrieval consistency metric. For each article in a client's corpus, we constructed a query from the article's title and ran it against the remaining corpus, excluding the query article itself. A hit was counted when at least one retrieved result scored ≥ 0.55, the same threshold used in production. This measures whether the embedding space is internally consistent - whether a corpus article's title reliably surfaces semantically similar content. This is not Recall@K in the strict information retrieval sense, which would require labeled relevance judgments, but a valid proxy given the absence of a labeled test set.

This evaluation was run once during development to determine the optimal alpha value across a grid of [0.3, 0.5, 0.7, 0.9, 0.95, 1.0]. Across all four clients, consistency scores increased monotonically with alpha, confirming that dense-only retrieval ($\alpha = 1.0$) outperformed every hybrid blend in every case. At $\alpha = 1.0$, Client 1 achieved the highest consistency at 0.875, followed by Client 2 at 0.847, Client 3 at 0.798, and Client 4 at 0.617. The notably lower score for Client 4 is consistent with that corpus having mixed topic coverage and heavy concentration in commercial real estate and employment law, which may produce less internal semantic diversity than the personal injury or healthcare corpora. These parameters are now fixed in the system configuration and applied uniformly to new clients by default; re-running the evaluation for a new client would require manually enabling the commented evaluation code, which is not currently automated.

5.2 Human Evaluation

While consistency measures technical retrieval quality, it does not guarantee that retrieved stories are actionable PR opportunities. To capture real-world relevance, we implemented a human evaluation loop:

1. Weekly digests are exported as Word documents containing all strong and adjacent matches
2. A PR practitioner reviews each match and marks whether it represents a meaningful opportunity
3. Ratings are logged in feedback_log.json, forming a structured record of human preferences

This inclusion is important because PR value is subjective. A retrieved match might be technically similar to the client's expertise but still not a story worth pitching. The human evaluation ensures the system surfaces stories that practitioners would actually act on. Over time, the system could leverage these preference pairs to fine-tune a reranking layer (DPO) to better reflect practitioner judgment.

6. Limitations and Future Work

While the current prototype demonstrates a strong foundation for pitch generation, several key limitations must be addressed to transition from a pilot to a production-ready system.

6.1 Core System Limitations

The system's effectiveness is currently constrained by API rate limits and corpus freshness. The Groq free tier imposes a cap of 100K tokens per day, which limits the volume of pitches generated per run and necessitates a fallback template for outages. Additionally, the system utilizes a static corpus that does not automatically update. In a live environment, monthly or quarterly re-indexing would be required to ensure the "expertise fingerprint" of each client remains current.

Data density also varies significantly across the portfolio. Client 2's corpus is the smallest (94 articles), requiring supplemental health calendar seeds to maintain performance. Conversely, Client 3's corpus has sparse topic labels, with the majority categorized as 'general.' Topic labels are not used in retrieval, as embeddings are built from article text only, but the lack of granular labels limits our ability to perform topic-level analysis of retrieval performance for that client.

6.2 Evaluation and Refinement Gaps

On the technical side, performance is currently measured using a retrieval consistency metric, which cannot fully capture real-world pitch relevance or labeled held-out generalization. Furthermore, DPO (Direct Preference Optimization) retraining has not yet been implemented. Although the infrastructure to collect feedback is live, the system cannot yet autonomously fine-tune its bi-encoder or train a cross-encoder reranker until enough preference pairs are accumulated.

Finally, while the architecture is designed to support seasonal triggers, specific pitch calendars for Client 1, Client 3, and Client 4 are currently absent. Integrating these, such as aligning Client 3 pitches with FCC regulatory cycles, represents a critical step toward achieving fully automated, context-aware outreach.

7. Path to Product

The agency that informed this project expressed interest in a deployable tool, not just a research prototype. Several components are already production-viable: the ChromaDB collections are persistent and portable, the RSS pipeline has no rate limits, Groq pitch generation produces PR-quality output when given a good corpus and relevant headlines, and a web-based digest UI has been built that renders strong and adjacent matches by client, displays generated pitches, and accepts match ratings from the PR practitioner.

The primary remaining gap between current state and a fully deployed product is automation and hosting. Today the system still requires running the Jupyter notebook to generate the digest JSON that powers the UI. A production deployment would replace this with a scheduled pipeline that runs automatically on a weekly cadence and pushes results to a hosted version of the interface, eliminating the need for any technical setup from the end user.

The multi-client architecture is already designed for agency use. Each client is an isolated collection with its own corpus, queries, and thresholds. Adding a new client requires uploading their article corpus, configuring topic queries, and running the indexing pipeline: a one-time 30-minute setup per client.

The feedback loop is the most strategically valuable component for a product. Every time a PR professional rates a match, the system learns what "good" looks like for that client. Over time, accumulated preference pairs could fine-tune a reranking layer specific to each client's sensibility, like the DPO pipeline applied to a real problem.

8. Conclusion

This system demonstrates that semantic retrieval and retrieval-augmented generation can be combined into a practical tool that solves a genuine problem identified through direct practitioner consultation.

The key technical finding is the superiority of dense retrieval over hybrid search for this corpus. Varied journalistic language across six years of placements means semantic similarity captures relevance that keyword matching misses entirely. This is not a failure of the hybrid design, but a meaningful empirical result that informs how the system should be configured and how future clients should be onboarded.

The most important practical insight is the distinction between what keyword alerts provide and what semantic matching provides. Keyword alerts surface stories that already mention the client's topics, often

competitor coverage or stories where the client has already been quoted. Semantic matching surfaces stories the client could enter: stories where they have relevant expertise but have not yet been quoted. That gap is where PR value is created, and it is exactly the gap this system is designed to close.

The multi-client architecture, Client 2's health calendar feature, human feedback loop, web-based digest UI, and path-to-product design extend the system toward something a real agency could deploy. The backend pipeline, retrieval system, pitch generation, and practitioner-facing interface are all in place; the next step is automating the weekly run and hosting the tool for production use.